Switching And VLANS

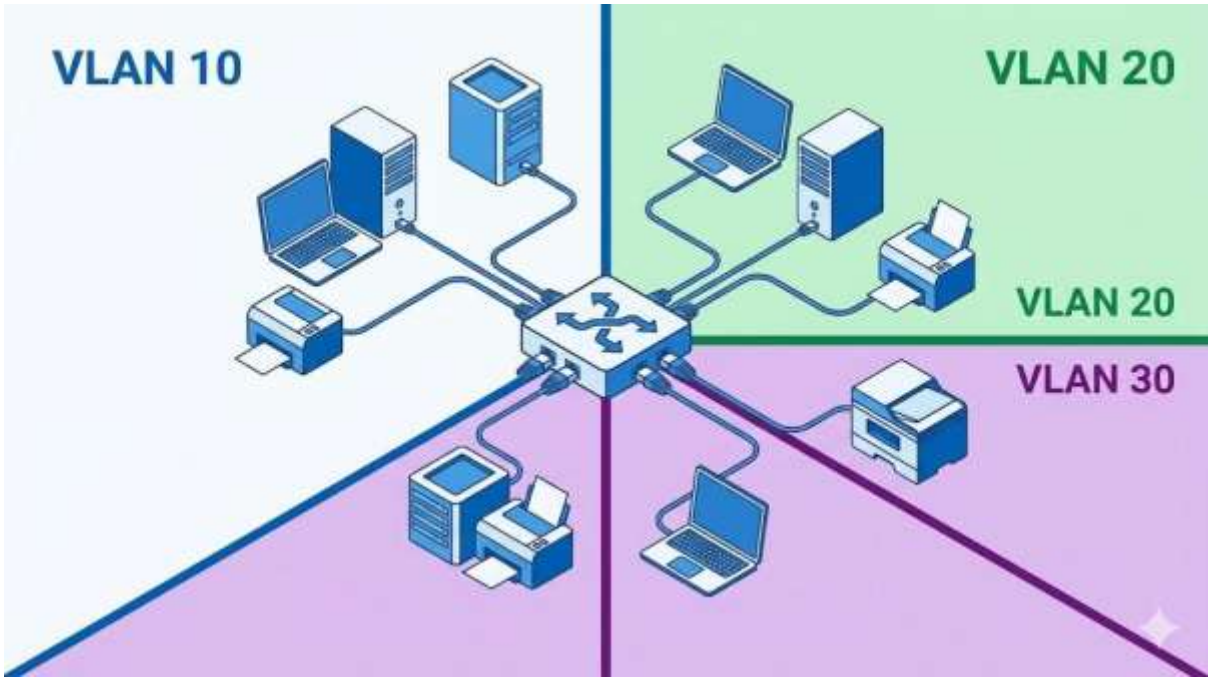
Tasks

1.Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

ANS:

Network Diagram – Devices Connected to a Switch Using VLANs

Aim: To show how three devices connect to a switch using different VLANs.



VLAN Table

Device	VLAN	Purpose
Smart TV	VLAN 10	Streaming
Laptop	VLAN 20	Personal use
Friend's Phone	VLAN 20	Personal use
Switch	VLAN 10 and 20	Connects all devices

Explanation (Easy Words)

1. Connect all three devices to the switch.
2. Smart TV is placed in VLAN 10.

3. Laptop and phone are placed in VLAN 20.
4. VLAN 10 is used for streaming.
5. VLAN 20 is used for personal devices.
6. VLANs separate devices into different networks.

Real-Life Use

In a home or office, VLANs help separate smart TVs, computers and personal devices for better network management and security.

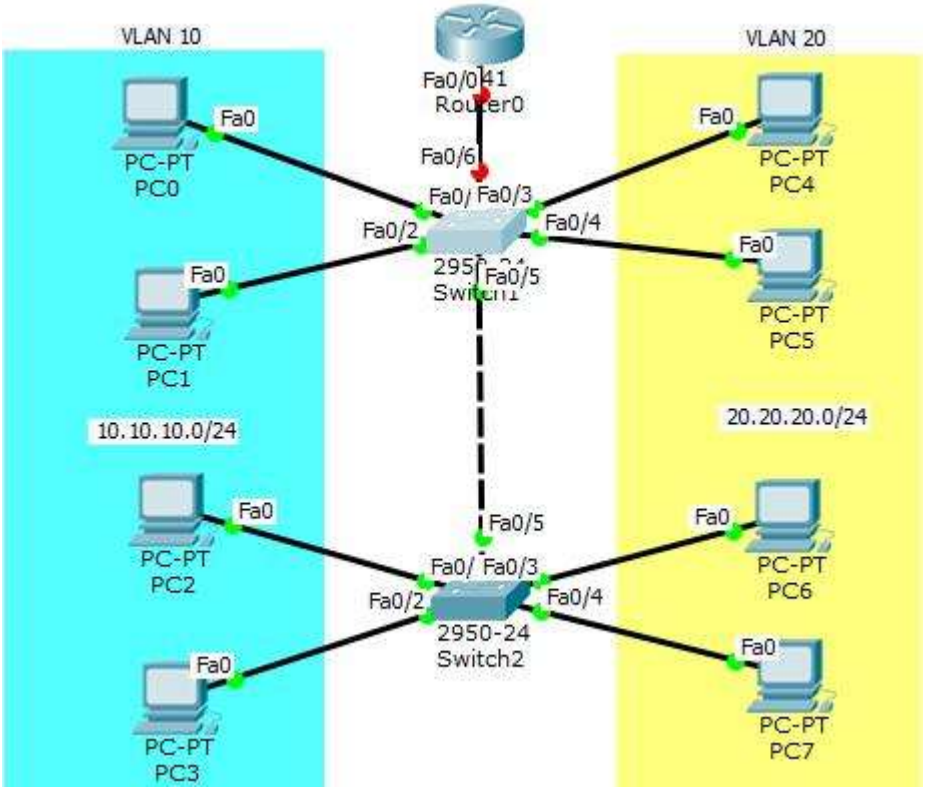
2.Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

ANS:

Configure VLAN 10 and VLAN 20 Using Cisco Packet Tracer

Aim: To create two VLANs and connect PCs to different VLANs.



Step 1: Create Network Topology

We need:

- 1 Switch
- 5 PCs
- VLAN 10 – 3 PCs
- VLAN 20 – 2 PCs

Step 2: Create VLANs

Open Switch → CLI and type:

enable

configure terminal

vlan 10

name Streaming

exit

vlan 20

name Personal

exit

Step 3: Assign PCs to VLANs

VLAN 10 – Three PCs

< / > CISCO

interface range fastEthernet 0/1-3

switchport mode access

switchport access vlan 10

exit

VLAN 20 – Two PCs

</> CISCO

interface range fastEthernet 0/4-5

switchport mode access

switchport access vlan 20

exit

Step 4: Assign IP Addresses

PC	VLAN	IP Address
PC1	10	192.168.10.1
PC2	10	192.168.10.2
PC3	10	192.168.10.3
PC4	20	192.168.20.1
PC5	20	192.168.20.2

Subnet mask for all PCs: 255.255.255.0

Step 5: Verify VLANs

On the switch CLI, type:

</ > CISCO

show vlan brief

You should see:

- VLAN 10 – Ports Fa0/1, Fa0/2, Fa0/3
- VLAN 20 – Ports Fa0/4, Fa0/5

Step 6: Test Ping

Test 1: PC1 to PC2

</> CMD

ping 192.168.10.2

Success

PCs in the same VLAN can communicate.

Test 2: PC1 to PC4

</> CMD

ping 192.168.20.1

Failed

PCs in different VLANs cannot communicate without a router or Layer 3 switch.

Real-Life Use

- VLAN 10 can be used for streaming devices.
- VLAN 20 can be used for personal devices.
- Offices use VLANs to separate employee and guest networks.

3.In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

ANS:

Move a PC from VLAN 10 to VLAN 20

Aim: To move one computer from VLAN 10 to VLAN 20 and check its connection.

Step 1: Before (VLAN 10)

Open Switch CLI and type:

```
enable
configure terminal
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit
```

PC1 is now in VLAN 10.

Step 2: Check VLAN

```
show vlan brief
```

Step 3: Move PC1 to VLAN 20

Type:

configure terminal

interface fastEthernet 0/1

switchport access vlan 20

exit

Now PC1 is in VLAN 20.

Step 4: Check Again

show vlan brief

You can see PC1's port in VLAN 20.

Step 5: Test Connection

PC	VLAN	IP Address
PC1	20	192.168.20.1
PC2	10	192.168.10.2
PC3	20	192.168.20.2

Test 1: Ping PC3.

ping 192.168.20.2

Result: Successful.

Test 2: Ping PC2.

ping 192.168.10.2

Result: Failed (without a router).

Real-Life Use

In an office, when an employee changes departments, the admin can move their computer to another VLAN.

4.Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

ANS:

Assign 10 Switch Ports to VLAN 30

Aim: To assign 10 switch ports to VLAN 30 using Cisco Packet Tracer.

Step 1: Open Switch CLI

Open Cisco Packet Tracer → Select Switch → CLI.

Step 2: Enter Commands

```
enable
configure terminal

vlan 30
name Office
exit

interface range fastEthernet 0/1-10
switchport mode access
switchport access vlan 30
exit
```

end

Step 3: Check Configuration

Type:

show vlan brief

You can see ports Fa0/1 to Fa0/10 in VLAN 30.

Explanation (Easy Words)

1. enable – Start admin mode.
2. configure terminal – Open settings.
3. vlan 30 – Create VLAN 30.
4. name Office – Give VLAN a name.
5. interface range fastEthernet 0/1-10 – Select 10 ports together.
6. switchport access vlan 30 – Assign all 10 ports to VLAN 30.
7. show vlan brief – Check the result.

Real-Life Use

In an office, an admin can put 10 employee computers into the same VLAN using one command.